MATH 381 Exam 2

December 5, 2025

Name (as on Gradescope): Solutions

Student #: _____

Problem:	1	2	Total
Points:	15	15	30

INSTRUCTIONS:

- You have 50 minutes to take the test.
- Write your solution below the problem. There is scratch paper at the back of the test.
- If you need extra space to write your solution, use the paper at the back and indicate on which page your solution continues.
- The test is double-sided. Make sure you are reading the backs of pages!
- Unless otherwise stated, show all your work for full credit.
- You are allowed to use one 8.5"×11" sheet of notes, front and back.
- Calculators are not allowed.

Good luck!

1. Consider the following game. A person is standing on a number line, at either 0, 1, 2, 3, or 4. Each step of the game, they do the following based on where they are currently standing.

   - If they are standing at 0, they always move to 1.
   - If they are standing at 4, they always move to 3.
   - If they are standing at 1, 2, or 3, they flip a fair coin. If heads, they move up by 1. If tails, they move down by 1. For example, if they are currently at 3 and flip tails, they would move to 2.

   Let X_0 be the person's initial position, and let X_t be the person's position after t steps. Then X_0, X_1, X_2, ... is a Markov chain.

   (a) (4 points) Write the transition probability matrix for this Markov chain. You should order the states in increasing order, that is, 0, 1, 2, 3, 4.

$$\begin{pmatrix} 0 & 1 & 0 & 0 & 0 \\ 1/2 & 0 & 1/2 & 0 & 0 \\ 0 & 1/2 & 0 & 1/2 & 0 \\ 0 & 0 & 1/2 & 0 & 1/2 \\ 0 & 0 & 0 & 1 & 0 \end{pmatrix}$$

   (b) (3 points) Which of the following terms describes this Markov chain: aperiodic, ergodic, irreducible? Write down all that apply. You do **not** need to explain your answer.

irreducible

(c) (4 points) The vector $(1, 2, 2, 2, 1)$ is a left eigenvector for the transition probability matrix. What is a stationary distribution for this Markov chain? Answer this question **without** solving a system of equations.

$1+2+2+2+1 = 8$

$\frac{1}{8}(1,2,2,2,1) = \boxed{\left(\frac{1}{8}, \frac{1}{4}, \frac{1}{4}, \frac{1}{4}, 1\right)}$

(d) (4 points) Let P be the previous transition probability matrix, with states ordered in increasing order. It is observed that

$$P^{100} \approx \begin{pmatrix} 0.25 & 0 & 0.5 & 0 & 0.25 \\ 0 & 0.5 & 0 & 0.5 & 0 \\ 0.25 & 0 & 0.5 & 0 & 0.25 \\ 0 & 0.5 & 0 & 0.5 & 0 \\ 0.25 & 0 & 0.5 & 0 & 0.25 \end{pmatrix} \qquad P^{101} \approx \begin{pmatrix} 0 & 0.5 & 0 & 0.5 & 0 \\ 0.25 & 0 & 0.5 & 0 & 0.25 \\ 0 & 0.5 & 0 & 0.5 & 0 \\ 0.25 & 0 & 0.5 & 0 & 0.25 \\ 0 & 0.5 & 0 & 0.5 & 0 \end{pmatrix}$$

(Here, P^{100} means P taken to the 100th power.) Interpret what the first row of P^{100} and the first row of P^{101} mean in terms of the game.

If the player starts at 0, after <u>100 steps</u> they will have 1/4 chance to be at 0, 1/2 chance to be at 2, and 1/4 chance to be at 4.

After <u>101 steps</u>, they will have 1/2 chance to be at 1, 1/2 chance to be at 3.

2. A festival has three tents, called A, B, and C, which contain people. The festival organizers want to model the movement of people at the festival in order to make better decisions. They decide to use the following Markov chain model: The state of the festival is stored as a triple (a, b, c), where a is the the number of people in tent A, b is the number of people in tent B, and c is the number of people in tent C. At each step of the chain,

- One of the **people** at the festival is chosen uniformly at random.
- Then, one of the tents is chosen uniformly at random.
- Then, the chosen person moves to the chosen tent. If the chosen person is already in the chosen tent, that person stays in that tent.

For example, the transition probability from state $(0, 3, 4)$ to state $(1, 2, 4)$ is $(3/7)(1/3) = 1/7$ because there is $3/7$ probability a person in tent B will be chosen, and $1/3$ probability that person will move to tent A.

(a) (2 points) Calculate the transition probability from state $(1, 2, 4)$ to state $(0, 3, 4)$.

$$\frac{1}{7} \cdot \frac{1}{3} = \boxed{\frac{1}{21}}$$

(b) (2 points) Calculate the transition probability from state $(1, 2, 4)$ to state $(1, 2, 4)$.

$$\frac{1}{7} \cdot \frac{1}{3} + \frac{2}{7} \cdot \frac{1}{3} + \frac{4}{7} \cdot \frac{1}{3} = \boxed{\frac{1}{3}}$$

(c) (3 points) Let M be the number of people in the tent(s) with the most people. (For example, if the state is $(0, 4, 3)$, then M is 4.) The organizers want to estimate the average of M in the long run. They do this by simulating the above Markov chain with initial state $(100, 0, 0)$ for 500 steps. They record the value of M at all 500 steps, and find the average of M over all 500 steps.

What is a way they could modify their code to make the result more likely to be accurate, **without** increasing the total number of steps?

Include a burn-in period (say of 100 steps) at the start, where we run the Markov chain but do not collect samples.

(d) (3 points) The organizers reason that if they run the simulation for long enough, then by the central limit theorem for independent identically distributed random variables, the average value of M over all steps should be approximately normally distributed with standard deviation $\sigma/\sqrt{n}$, where σ is the standard deviation of M and n is the number of steps. What is wrong with this reasoning? You need to explain what is wrong with their reasoning, **not** what the correct statement is.

The samples are <u>not independent</u>, since each sample is correlated to the previous one. Therefore this form of the CLT cannot be used.

Remark: After a long period of time, the average value M will still be normally distributed due to the CLT for ergodic chains. However, the standard deviation will be different from $\sigma/\sqrt{n}$.

PROBLEM 2 CONTINUED ON NEXT PAGE.

(e) (5 points) After observing the simulation, they decide that the distribution of states in the simulation does not match their real-life experience. They decide to modify the previous Markov chain using the Metropolis-Hastings algorithm so that the resulting Metropolis chain has stationary distribution proportional to the function F, where

$$F(a, b, c) = a^2 + b^2 + c^2.$$

Suppose that in the Metropolis chain, the proposal chain (i.e. the original chain) proposes the transition $(0, 3, 4)$ to $(1, 2, 4)$. What is the probability that the Metropolis chain accepts this transition?

$$\alpha = \frac{F(1,2,4)}{F(0,3,4)} \cdot \frac{P_{124,034}}{P_{034,124}}$$

$$= \frac{21}{25} \cdot \frac{1/21}{1/7} \quad \leftarrow \text{from part (a)}$$

$$= \boxed{\frac{7}{25}}$$

END OF EXAM.